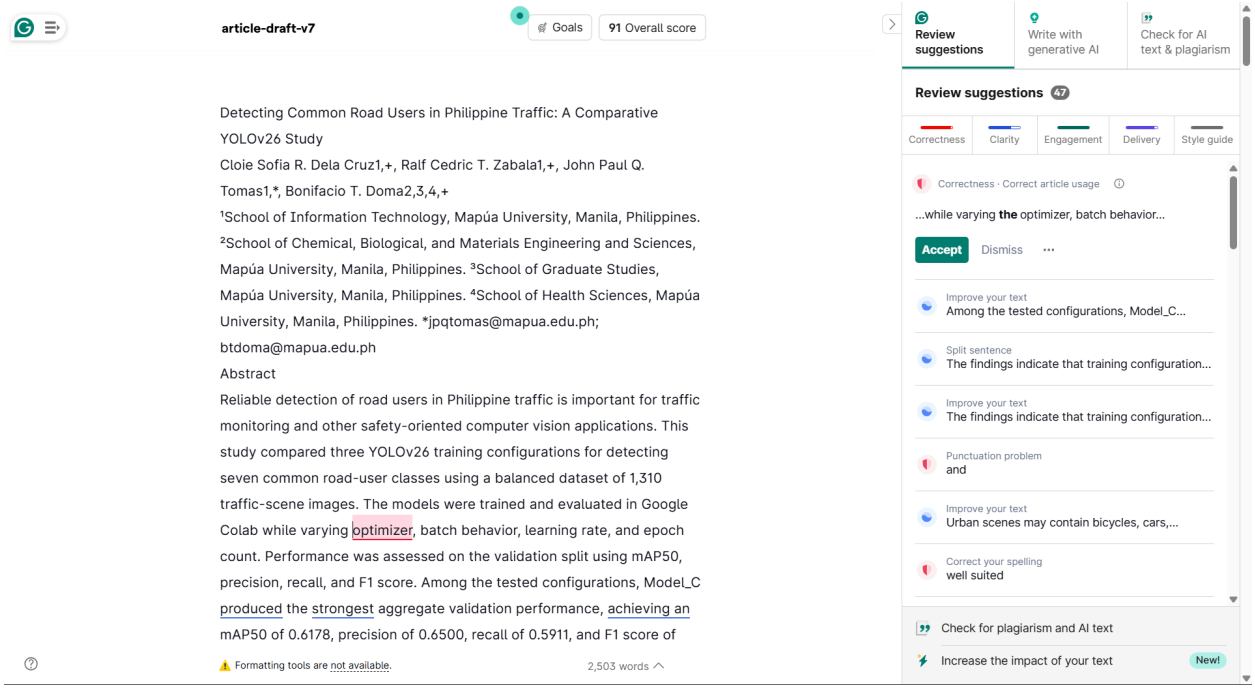
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Detecting Common Road Users in Philippine Traffic: A Comparative YOLOv26 Study

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Abstract

Reliable detection of road users in Philippine traffic is important for traffic monitoring and other safety-oriented computer vision applications. This study compared three YOLOv26 training configurations for detecting seven common road-user classes using a balanced dataset of 1,310 traffic-scene images. The models were trained and evaluated in Google Colab while varying optimizer, batch behavior, learning rate, and epoch count. Performance was assessed on the validation split using mAP50, precision, recall, and F1 score. Among the tested configurations, Model_C produced the strongest aggregate validation performance, achieving an mAP50 of 0.6178, precision of 0.6500, recall of 0.5911, and F1 score of 0.6192. Class-wise results remained uneven: Jeepney, SUV, and Truck were comparatively strong, whereas Tricycle was persistently difficult across all runs. The findings indicate that training configuration materially affects detection quality on this dataset, but the current comparison should be interpreted as a comparative validation study rather than a final deployment benchmark because multiple hyperparameters changed simultaneously and the held-out test split was not yet used for the final model selection.

Introduction

Road-user detection is a foundational computer-vision task because it supports scene understanding, surveillance, tracking, and intelligent transportation analysis [1,2]. In traffic environments, this task is especially challenging because road scenes are crowded, dynamic, and affected by occlusion, scale variation, illumination changes, and camera-related distortion [3-5][3,4,5]. The problem becomes more difficult when several road-user categories appear in the same frame and differ substantially in size, appearance, and frequency.

The Philippine traffic context provides a meaningful setting for this problem. Urban scenes may contain bicycles, cars, jeepneys, pedestrians, sport utility vehicles, tricycles, and trucks within the same view. This

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